

From what is known to what is done: The science of innovation, implementation, and sustainment

Simone J. Skeen, MA^{1,2,3}



1. Department of Social, Behavioral, and Population Sciences;
2. Center for Community-Engaged Artificial Intelligence, School of Public Health and Tropical Medicine, Tulane University
3. Department of Psychology, Hunter College, City University of New York

Part I
Diffusion of
Innovation: Drivers
of adoption

Part II
Implementation
Science: The
CFIR and its
applications

PART I

Diffusion of Innovation: Drivers of adoption

You've just designed and trialed
the *perfect intervention*.

Its efficacy surpasses that of all
others.

Now what?

You've just designed and trialed
the *perfect intervention*.

Its efficacy surpasses that of all
others.

Now what?

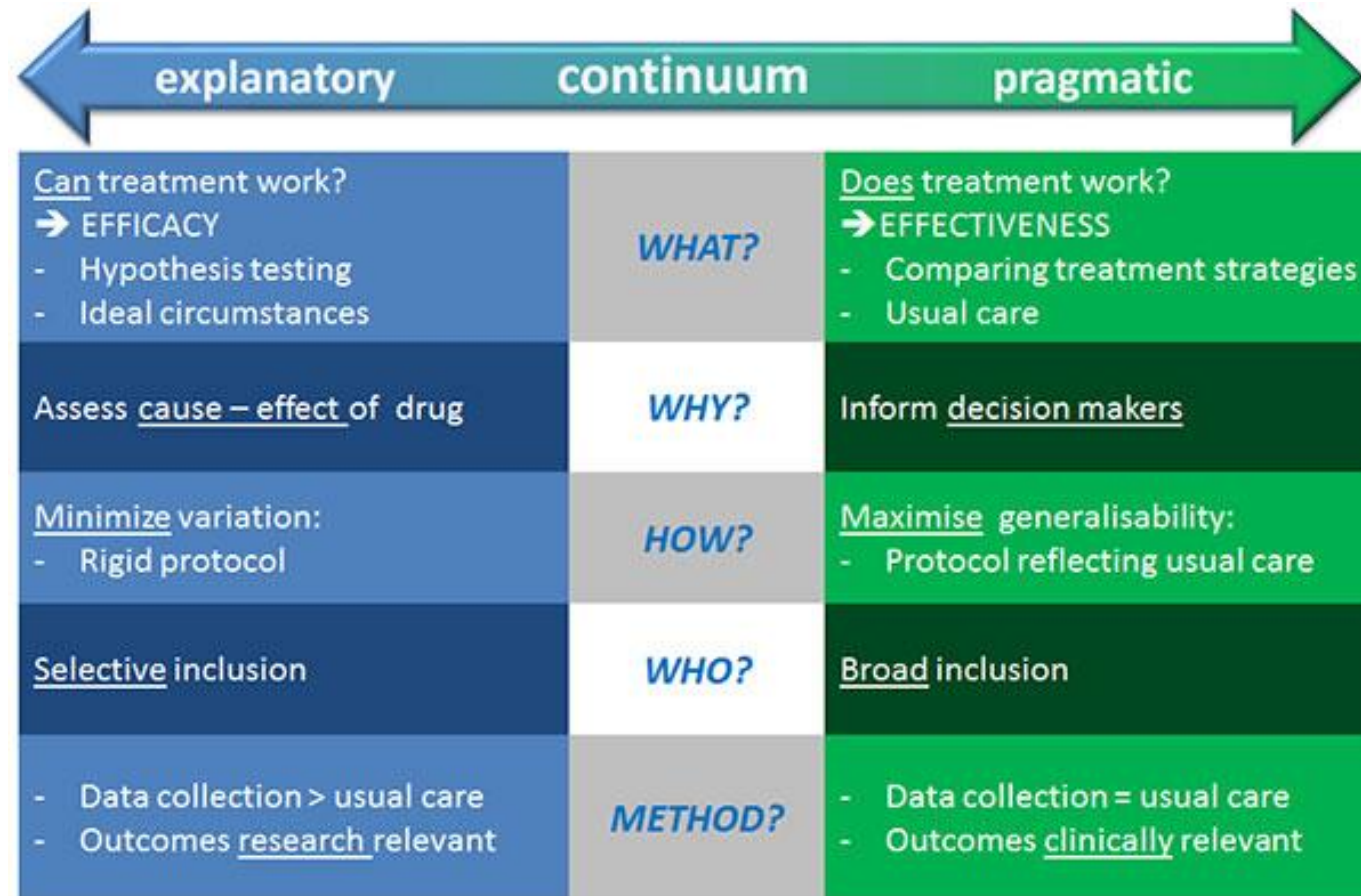


Fig. 1. Continuum between explanatory and pragmatic trials
(RWE Navigator, n.d.).

How do we bridge the **research–practice gap**?

Diffusion $\neq$ **Dissemination**

Diffusion

The passive, untargeted, unplanned, and uncontrolled spread of new interventions (Lomas, 1993; MacLean, 1996).

Dissemination

An active approach of spreading evidence-based interventions to the target audience via determined channels using planned strategies (Lomas, 1993; MacLean, 1996).

What are we trying to *spread*?

EBIs: Evidence-based interventions

Interventions with demonstrated ***efficacy*** and ***effectiveness*** e.g.

- Policies
- Programs
- Practice guidelines
- Screeners, diagnostic tools
- Instruments, devices
- Software

We want to see our interventions in use within the contexts in which they might make an **impact**.

As with any other *sustainable behavior change*, we have **theories to guide us**.



Fig. 2. The CDC's CEBI for HIV Prevention and One Mind PsyberGuide, two EBI dissemination platforms.

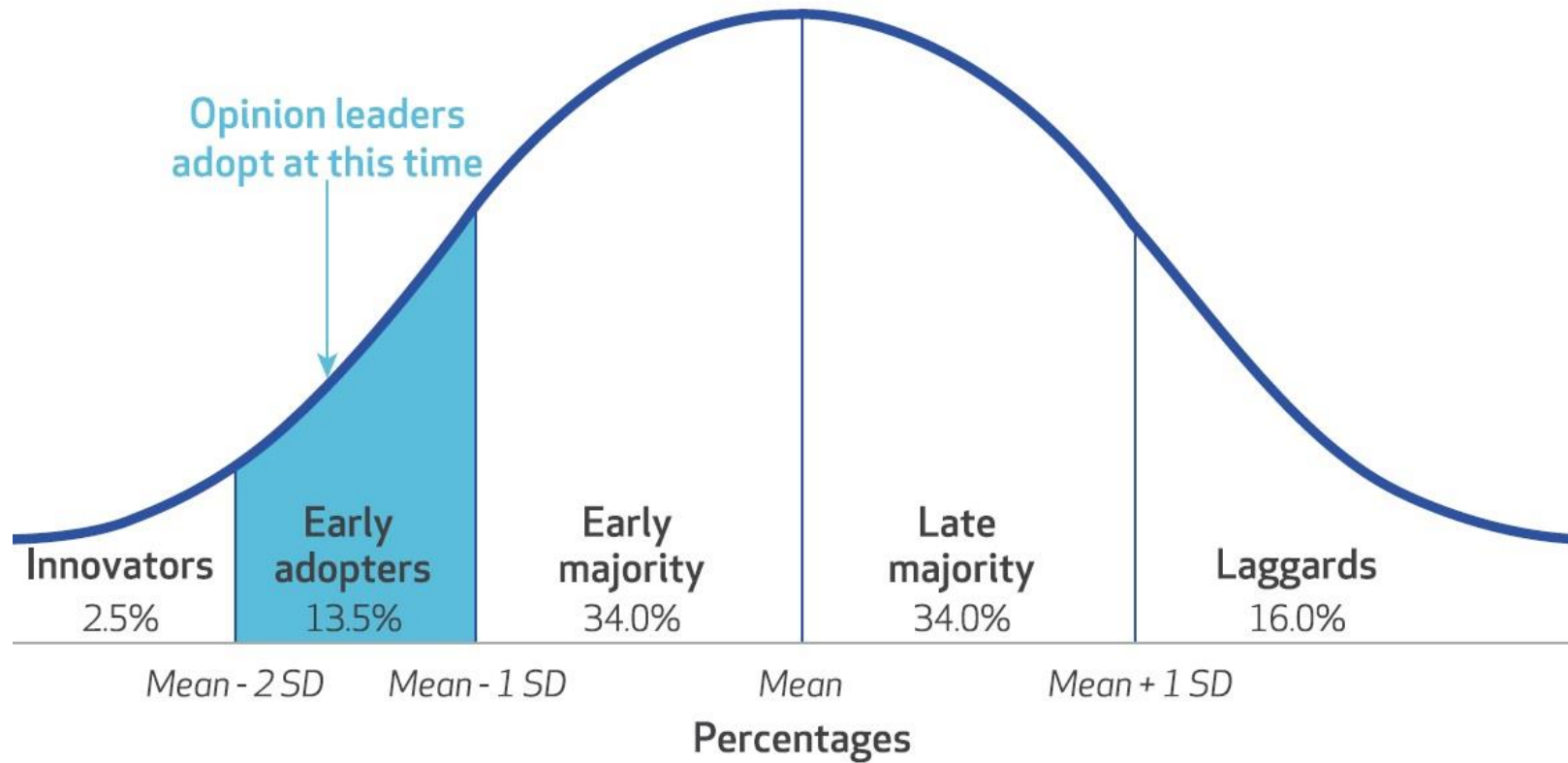


Fig. 3. Distribution of adopter innovativeness based on time of adoption (Dearing & Cox, 2018).

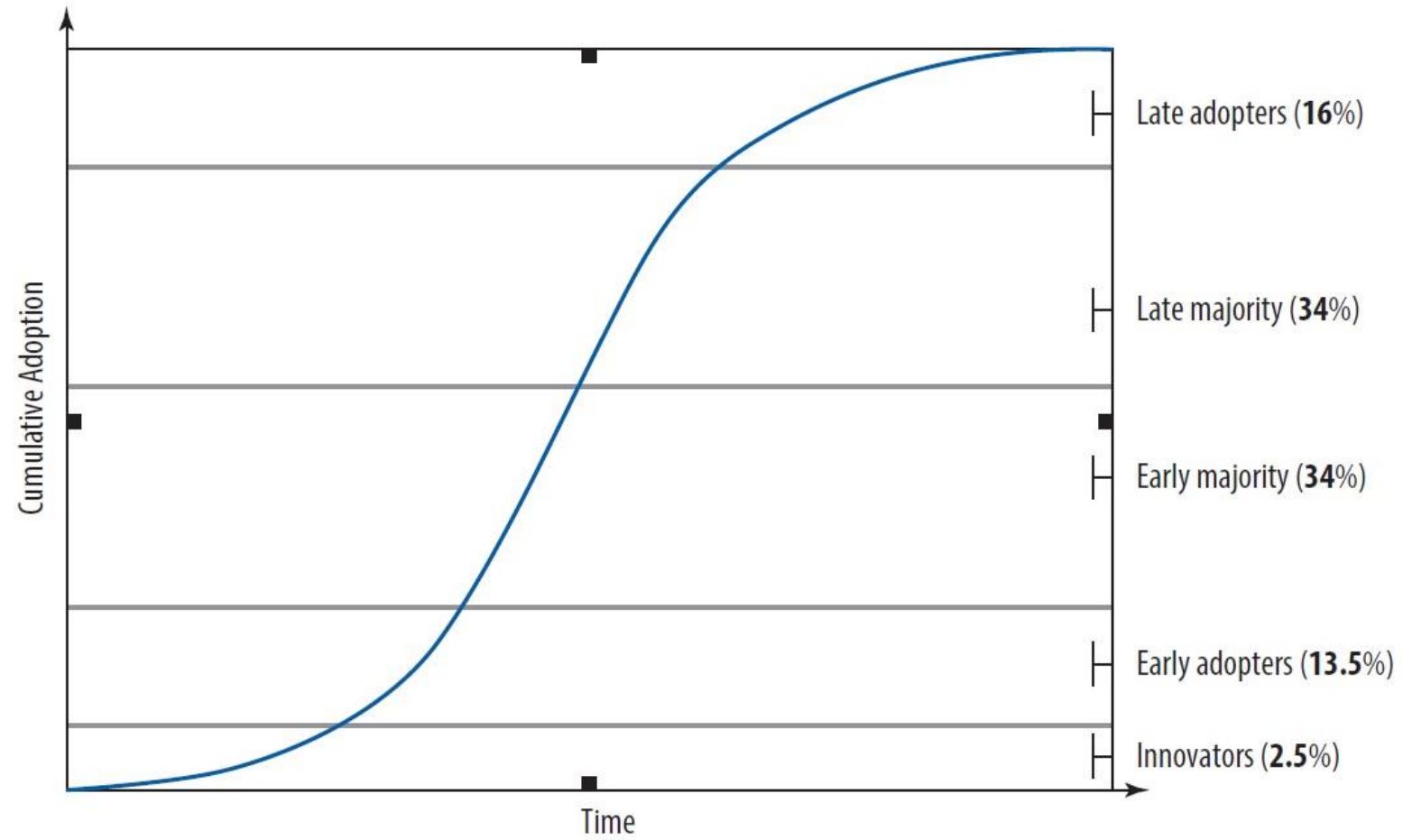


Fig. 4. S curve of diffusion (Brownson et al., 2015).

Diffusion of Innovations recognizes an **ordered stage model**.

Adopters can be conceptualized as *progressing* through these 5 stages (Brownson et al., 2015; Rogers, 2003).

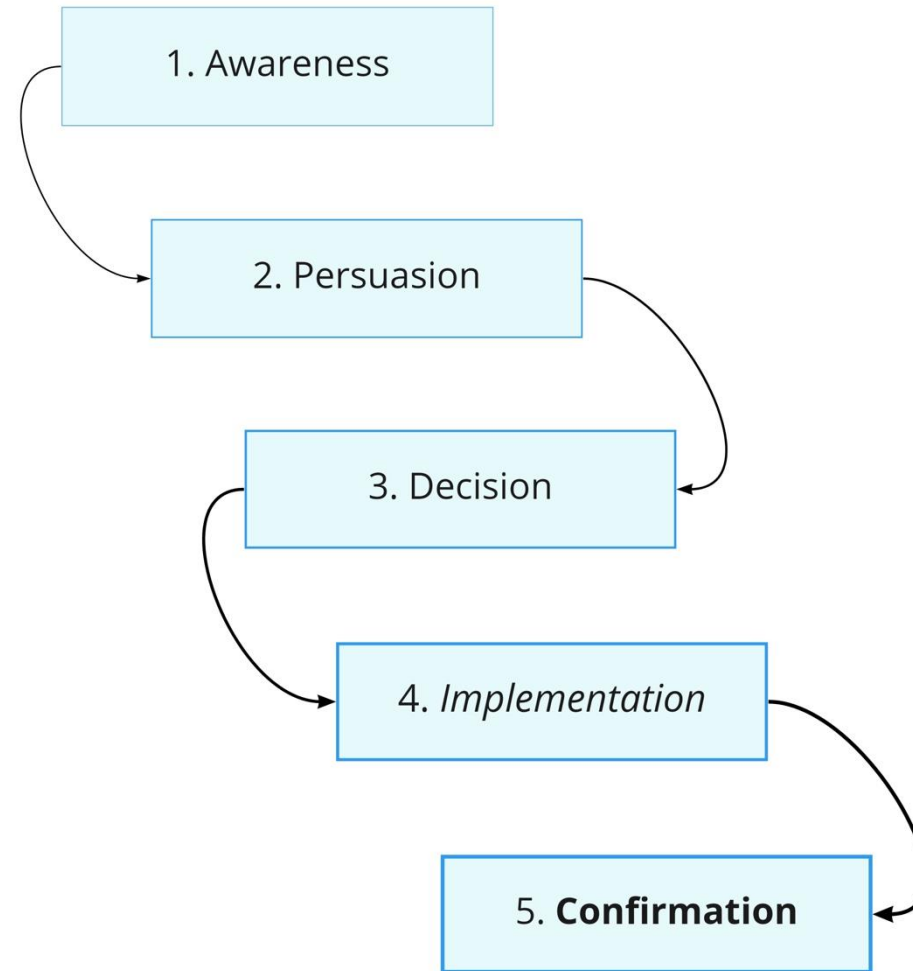


Fig. 5. Diffusion of Innovations: Stage-ordered model

The stage at which an adopter...

1. Awareness

Is **exposed** to an innovation, understands its value proposition.

2. Persuasion

Forms an attitude toward the innovation.

3. Decision

Forms a choice as **whether to adopt or reject** an innovation.

The stage at which an adopter...

4. *Implementation*

Puts an innovation to use; typically involves overt behavior change.

5. **Confirmation**

Seeks reinforcement for a decision already made.

Alternate stage models cover similar ground.

EPIS: Exploration, Preparation, Implementation, Sustainment, is common in mental and behavioral healthcare (Aarons et al., 2011).

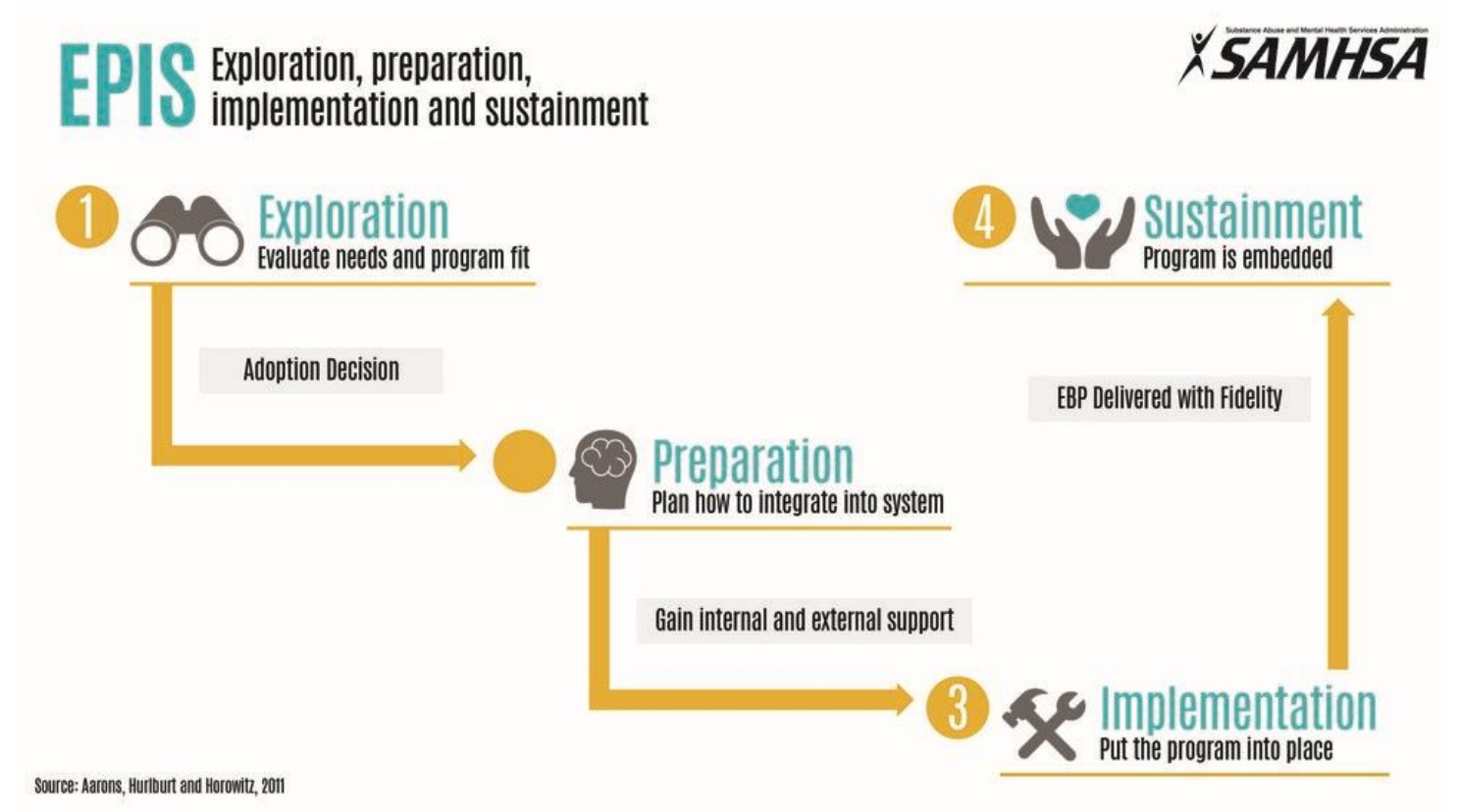


Fig. 6. The EPIS cycle – one depiction (SAMHSA, 2021).

FRONTLINE

FILMS PODCASTS TOPICS ▼ SCHEDULE

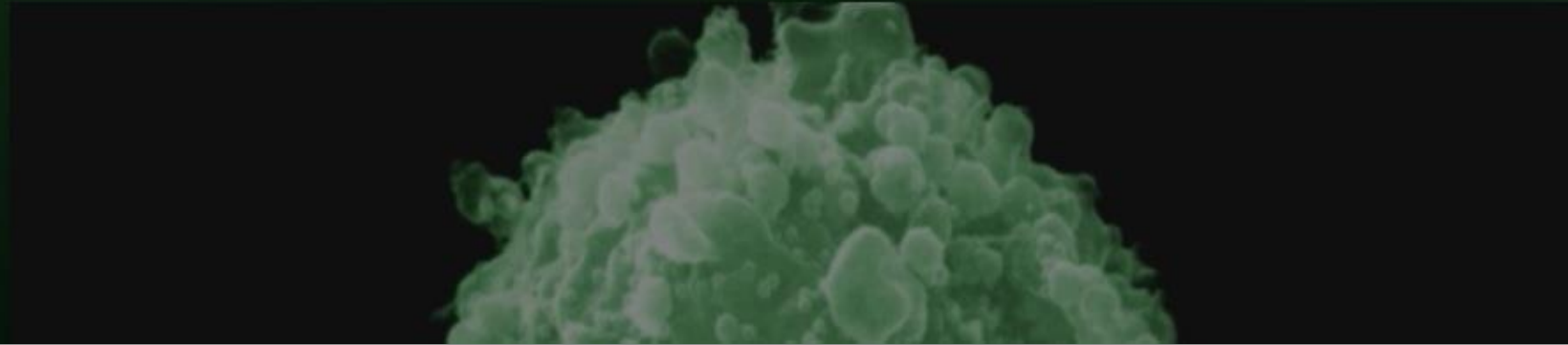
THE AGE OF AIDS

► PART I

May. 30, 2006 / 1h 53m

PART II

Apr. 09, 2022 / 1h 54m



<https://www.youtube.com/watch?v=KdTiPGVZNes>

["Setbacks" ~1:26:00–1:30:00]

Discussion

Through the lens of
Diffusion of Innovation:

What did we just watch?

Who did we just watch?

Where, along the adoption
process, is the UK? The
U.S.?

Inside a safe injection site that breaks the law but prevents overdoses

Toronto activists started an illegal safe injection site to prevent opioid overdoses.



By [VICE News](#)

October 3, 2017, 6:33am



[Share](#)



[Tweet](#)



[Snap](#)

https://www.youtube.com/watch?v=eF_mPzwKMfM

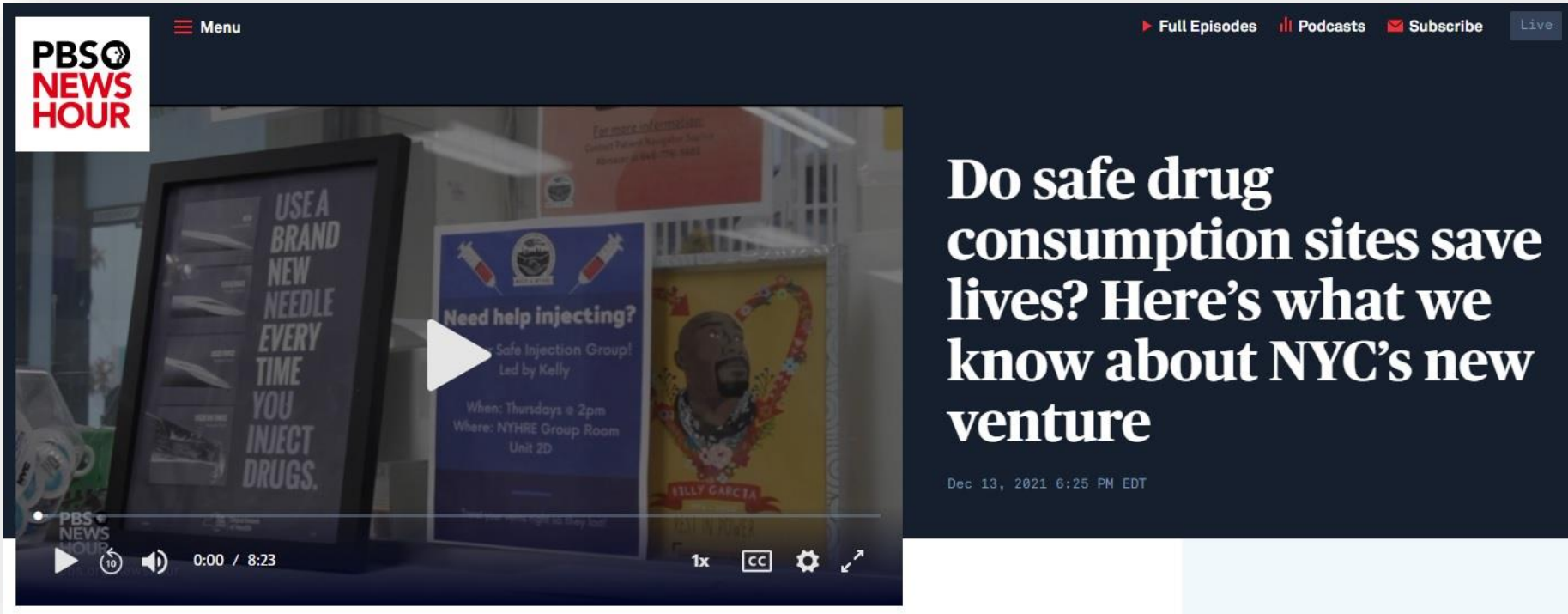
[0:00–2:17]

Discussion

Through the lens of Diffusion of Innovation:

Where, on the S-Curve of Diffusion, are we now?

Who, among stakeholders, is now persuaded?



<https://www.youtube.com/watch?v=JLnfnDLuDgk>
[0:00-5:09]

Discussion

Through the lens of Diffusion of Innovation:

Which stage have we reached in the diffusion process?

What differs from the first (Frontline) video, ~25 years earlier?

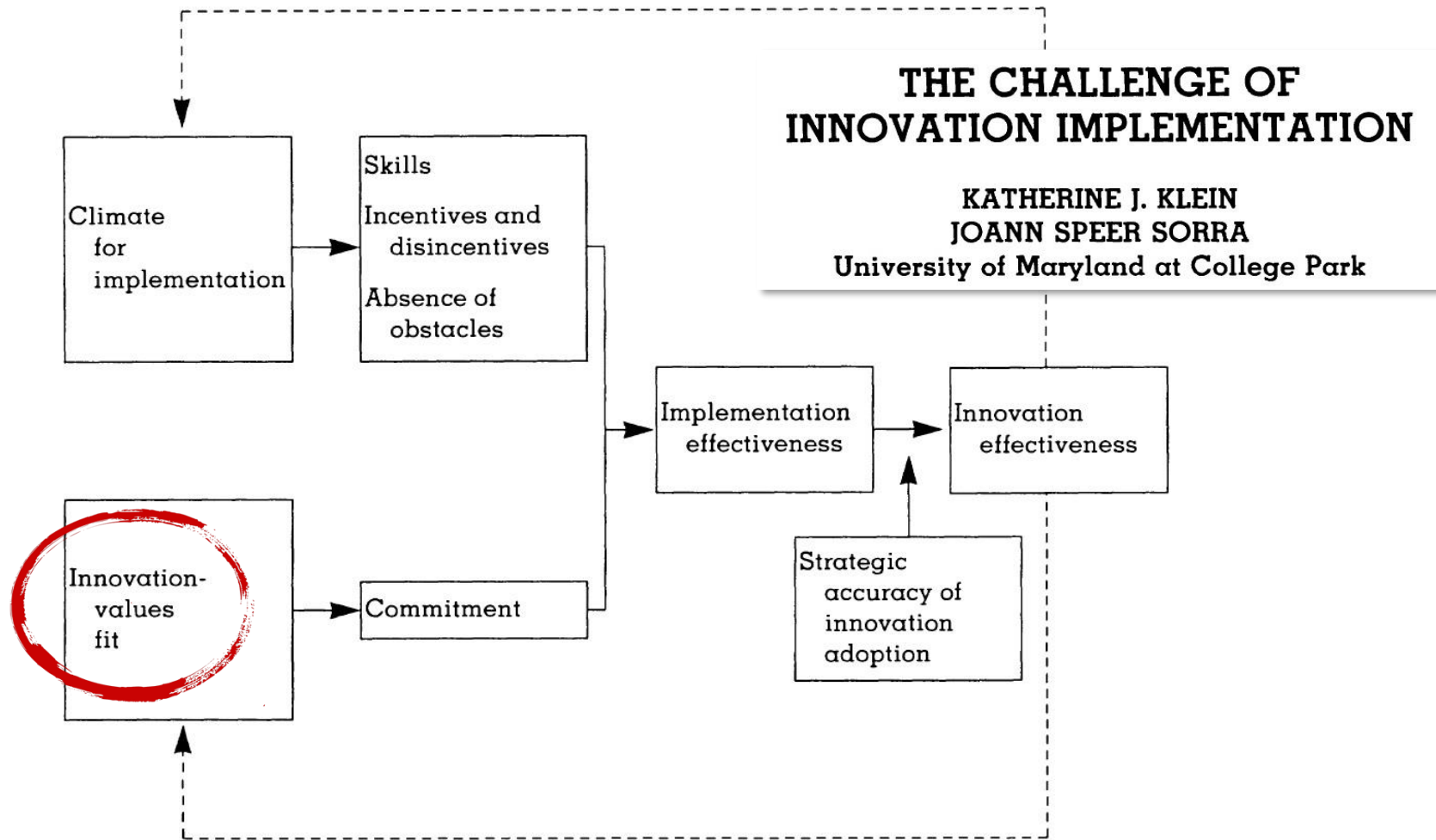


Fig. 7. Determinants and consequences of implementation effectiveness (Klein & Sorra, 1996).

INNOVATION-VALUES FIT

Adoption of an innovation is a Fx of the *perceived fit of the innovation* with an adopter's **values**.

Individuals, organizations, communities, societies, all have *stable* – but *not fixed* – values.

Innovation-Values Fit

The extent to which adopters perceive that use of an innovation will foster (or, conversely, inhibit) the fulfillment of their values (Klein & Sorra, 1996).



Fig. 8. Stevie the robot entertains a group of older adults (Savage, 2022).

DRAVE NEW WORLD DEPT. MAY 31, 2021 ISSUE

WHAT ROBOTS CAN— AND CAN'T—DO FOR THE OLD AND LONELY

Alberti writes that, for many of us, the loneliness of old people is held up as evidence of a lost era—of a better, kinder, more neighborly society gone by. For others, like some medical researchers, loneliness is a biological inevitability, a hazard of aging. But both formulations, Alberti argues, overlook the structures and the systems that have given rise to lonely people: industrialization, secularism, modernity. Some critics fear that, as social robots improve, they will be used as a means of care rationing—and that insisting on human company, at personal or family or communal expense, will be seen as a kind of indulgence.



Fig. 8. Stevie the robot entertains a group of older adults (Savage, 2022).

What is the **innovation**?

What (implicitly or explicitly) are our **values**?

Are they congruent?

PART II

Implementation Science: The CFIR and its applications

The critical link in the connection between research and practice is the **translation of EBIs** into broader application and impact (Aarons et al, 2011).

Implementation Science

Implementation science seeks to apply and integrate research evidence into practice and policy, with the goal of *closing the gap between what is known and what is done*

(U.S. Department of Veterans Affairs).

How do we assure that our interventions **generalize**?

Fidelity $\neq$ Drift

Fidelity

The extent to which an intervention is delivered as intended by the program developers and *in line with the program model*

(Breitenstein et al., 2010).

Drift

Non-systematic, unplanned, departures from the program model due to individual practitioner and contextual factors.

Discussion

Why might an intervention that is shown to be efficacious in a controlled setting fail in the real world?

Implementation Science



Research article

Open Access

Fostering implementation of health services research findings into practice: a consolidated framework for advancing implementation science

Laura J Damschroder*¹, David C Aron², Rosalind E Keith¹, Susan R Kirsh², Jeffery A Alexander³ and Julie C Lowery¹

Conclusion: The CFIR provides a pragmatic structure for approaching complex, interacting, multi-level, and transient states of constructs in the real world by embracing, consolidating, and unifying key constructs from published implementation theories. It can be used to guide formative evaluations and build the implementation knowledge base across multiple studies and settings.

Code	Topic/Description	Greenhalgh et al. ¹³	Klein, Conn & Sora ^{14,17}	Pedgrew & Whipp ¹⁸	Lewin ¹⁴	PARIS Model ¹⁵	Ofcove Model ¹⁶	Simpson ¹¹	Kochner & Yano ¹²	Stetler ²⁰	Edmondson et al. ¹¹	Kilbourne et al.	VanDeusen Lukas et al. ¹⁰	Grol et al. ^{2,18}	Mendel et al. ²	Flecken et al. ¹³	Brach et al. ¹⁸	Glisson et al. ^{10,12}	Feldstein & Glasgow ¹²	Frambach & Schillewaert ²¹
I. INTERVENTION CHARACTERISTICS																				
A	Intervention Source			✓						✓	✓			✓						
B	Evidence Strength & Quality				✓	✓	✓			✓	✓	✓		✓	✓		✓	✓	✓	✓
C	Relative advantage	✓				✓				✓	✓	✓		✓	✓		✓	✓	✓	✓
D	Adaptability	✓	✓	✓	✓	✓				✓	✓	✓		✓	✓		✓	✓	✓	✓
E	Triability	✓	✓		✓	✓				✓	✓	✓		✓	✓		✓	✓	✓	✓
F	Complexity	✓	✓							✓	✓	✓		✓	✓		✓	✓	✓	✓
G	Design Quality and Packaging		✓				✓				✓	✓		✓	✓		✓	✓	✓	✓
H	Cost													✓	✓		✓	✓	✓	✓
II. OUTER SETTING																				
A	Patient Needs & Resources			✓		✓	✓				✓	✓		✓	✓		✓	✓	✓	✓
B	Cosmopolitanism	✓		✓				✓			✓	✓		✓	✓		✓	✓	✓	✓
C	Peer Pressure										✓	✓		✓	✓		✓	✓	✓	✓
D	External Policies & Incentives	✓									✓	✓		✓	✓		✓	✓	✓	✓
III. INNER SETTING																				
A	Structural Characteristics	✓		✓								✓		✓	✓		✓	✓	✓	✓
B	Networks & Communications	✓	✓	✓		✓	✓	✓				✓		✓	✓		✓	✓	✓	✓
C	Culture	✓	✓	✓		✓	✓	✓						✓	✓		✓	✓	✓	✓
D	Implementation Climate			✓										✓	✓		✓	✓	✓	✓
1	Tension for Change	✓	✓	✓					✓	✓	✓	✓		✓	✓		✓	✓	✓	✓
2	Compatibility	✓	✓	✓				✓	✓	✓	✓	✓		✓	✓		✓	✓	✓	✓
3	Relative Priority			✓						✓	✓	✓		✓	✓		✓	✓	✓	✓
4	Organizational Incentives & Rewards	✓	✓	✓	✓	✓		✓	✓		✓	✓		✓	✓		✓	✓	✓	✓
5	Goals and Feedback	✓	✓	✓		✓		✓	✓		✓	✓		✓	✓		✓	✓	✓	✓
6	Learning Climate	✓	✓	✓				✓	✓		✓	✓		✓	✓		✓	✓	✓	✓
D	Readiness for Implementation			✓										✓	✓		✓	✓	✓	✓
1	Leadership Engagement	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓		✓	✓	✓	✓
2	Available Resources	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓		✓	✓	✓	✓
3	Access to knowledge and information	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓		✓	✓	✓	✓
IV. CHARACTERISTICS OF INDIVIDUALS																				
A	Knowledge & Beliefs about the Intervention	✓										✓		✓	✓		✓	✓	✓	✓
B	Self-efficacy	✓	✓	✓										✓	✓		✓	✓	✓	✓
C	Individual Stage of Change	✓	✓	✓										✓	✓		✓	✓	✓	✓
D	Individual Identification with Organization													✓	✓		✓	✓	✓	✓
E	Other Personal Attributes												✓	✓	✓		✓	✓	✓	✓
V. PROCESS																				
A	Planning	✓		✓		✓				✓	✓	✓		✓	✓	✓	✓	✓	✓	✓
B	Engaging									✓	✓	✓		✓	✓	✓	✓	✓	✓	✓
1	Opinion Leaders	✓				✓					✓	✓		✓	✓		✓	✓	✓	✓
2	Formally appointed internal implementation leaders			✓													✓	✓	✓	✓
3	Champions	✓										✓					✓	✓	✓	✓
4	External Change Agents	✓												✓	✓		✓	✓	✓	✓
C	Executing			✓						✓	✓	✓		✓	✓		✓	✓	✓	✓
D	Reflecting & Evaluating	✓		✓						✓	✓	✓		✓	✓		✓	✓	✓	✓

Fig. 8. Matrix of constructs from models in the literature to CFIR constructs (Damschroder et al., 2009).

CFIR: Consolidated Framework for Implementation Research

Represents a careful synthesis of extant theories of innovation, implementation, diffusion, and dissemination (Damschroder et al., 2009).

Including Diffusion of Innovations (Brownson et al., 2015).

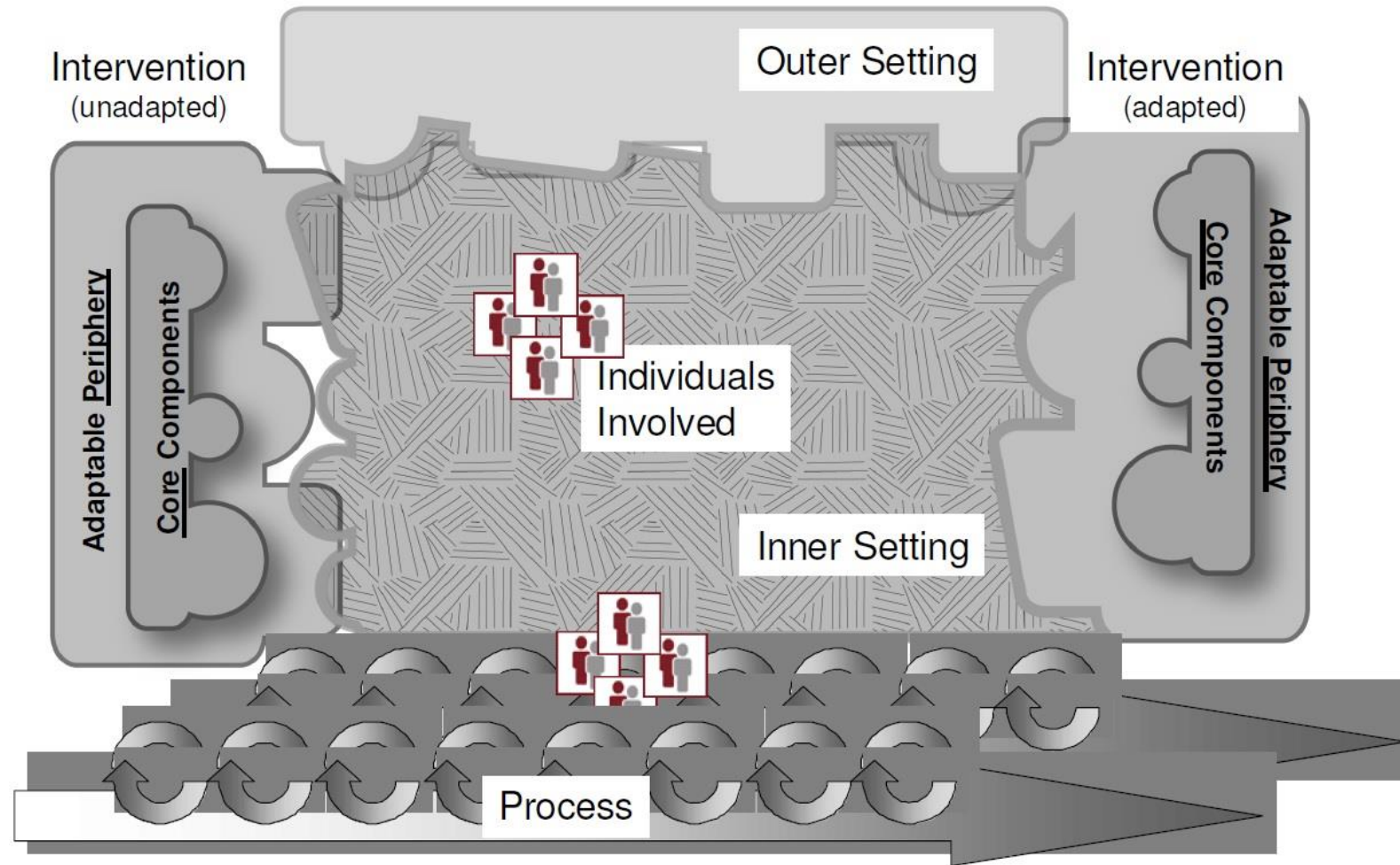


Fig. 9. Major Domains of the CFIR (Damschroder et al., 2009).

Consolidated Framework for Implementation Research (CFIR) 2.0

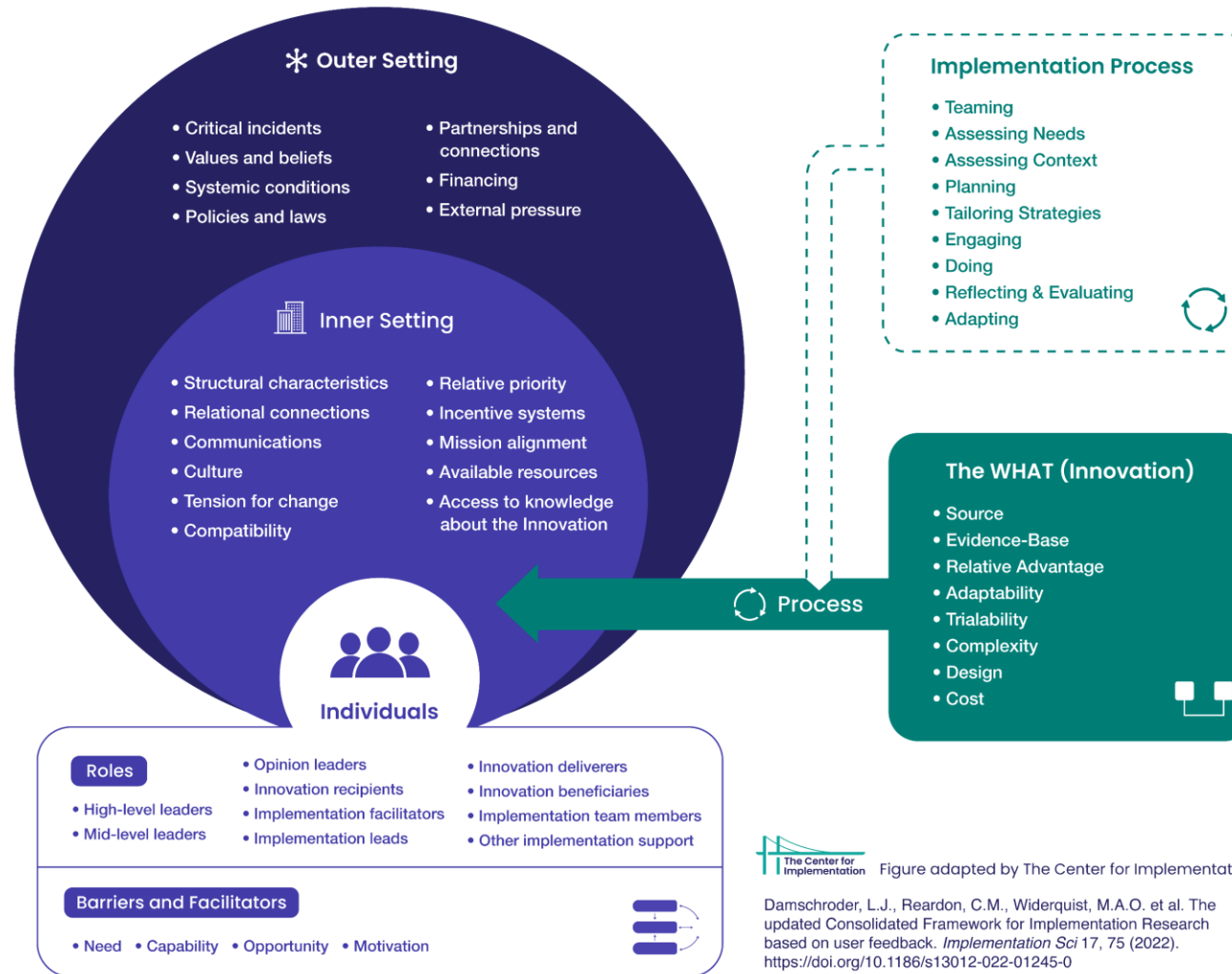


Figure adapted by The Center for Implementation

Damschroder, L.J., Reardon, C.M., Widerquist, M.A.O. et al. The updated Consolidated Framework for Implementation Research based on user feedback. *Implementation Sci* 17, 75 (2022). <https://doi.org/10.1186/s13012-022-01245-0>

Fig. 9. Major Domains of the CFIR 2.0 (Damschroder et al., 2022).

I. INTERVENTION CHARACTERISTICS		✓			
A	Intervention Source				
B	Evidence Strength & Quality				✓
C	Relative advantage	✓			
D	Adaptability	✓	✓	✓	✓
E	Trialability	✓	✓		✓
F	Complexity	✓	✓		
G	Design Quality and Packaging		✓		
H	Cost				
II. OUTER SETTING		✓			
A	Patient Needs & Resources				
B	Cosmopolitanism	✓		✓	
C	Peer Pressure	✓		✓	
D	External Policies & Incentives	✓		✓	

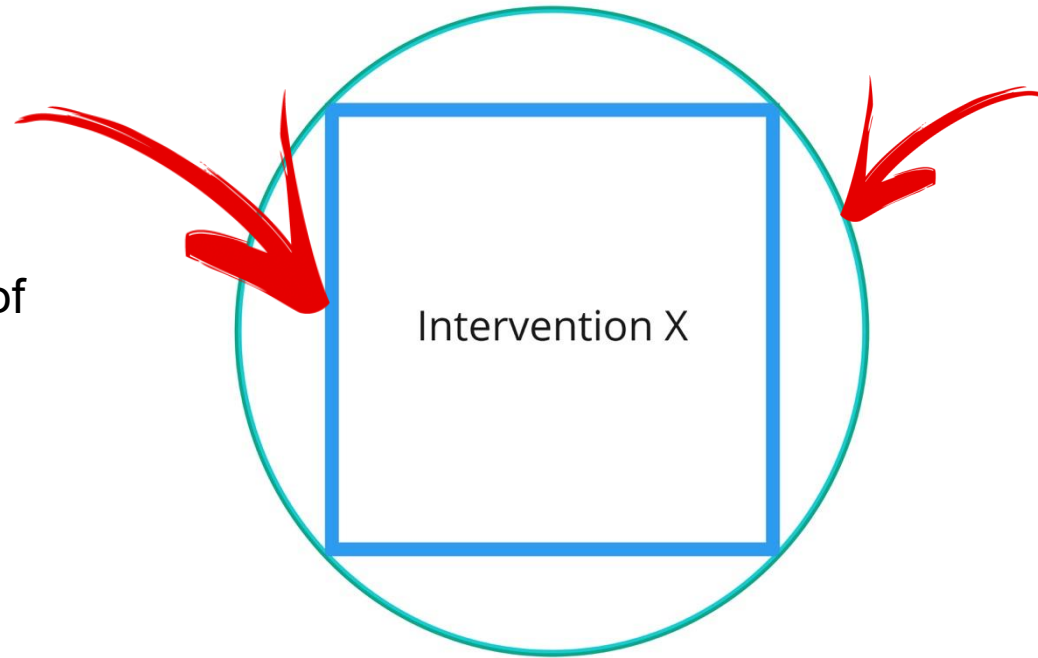
Fig. 9. Matrix of CFIR constructs: detail (Damschroder et al., 2009).

III. INNER SETTING					✓
A	Structural Characteristics	✓			✓
B	Networks & Communications	✓	✓		✓
C	Culture	✓	✓		✓
D	Implementation Climate				✓
1	Tension for Change	✓			✓
2	Compatibility	✓	✓		
3	Relative Priority		✓		
4	Organizational Incentives & Rewards		✓	✓	✓
5	Goals and Feedback	✓		✓	
6	Learning Climate	✓	✓		
D	Readiness for Implementation				✓
1	Leadership Engagement	✓	✓	✓	✓
2	Available Resources	✓	✓	✓	
3	Access to knowledge and information	✓	✓		✓

Fig. 10. Matrix of CFIR constructs: detail (Damschroder et al., 2009).

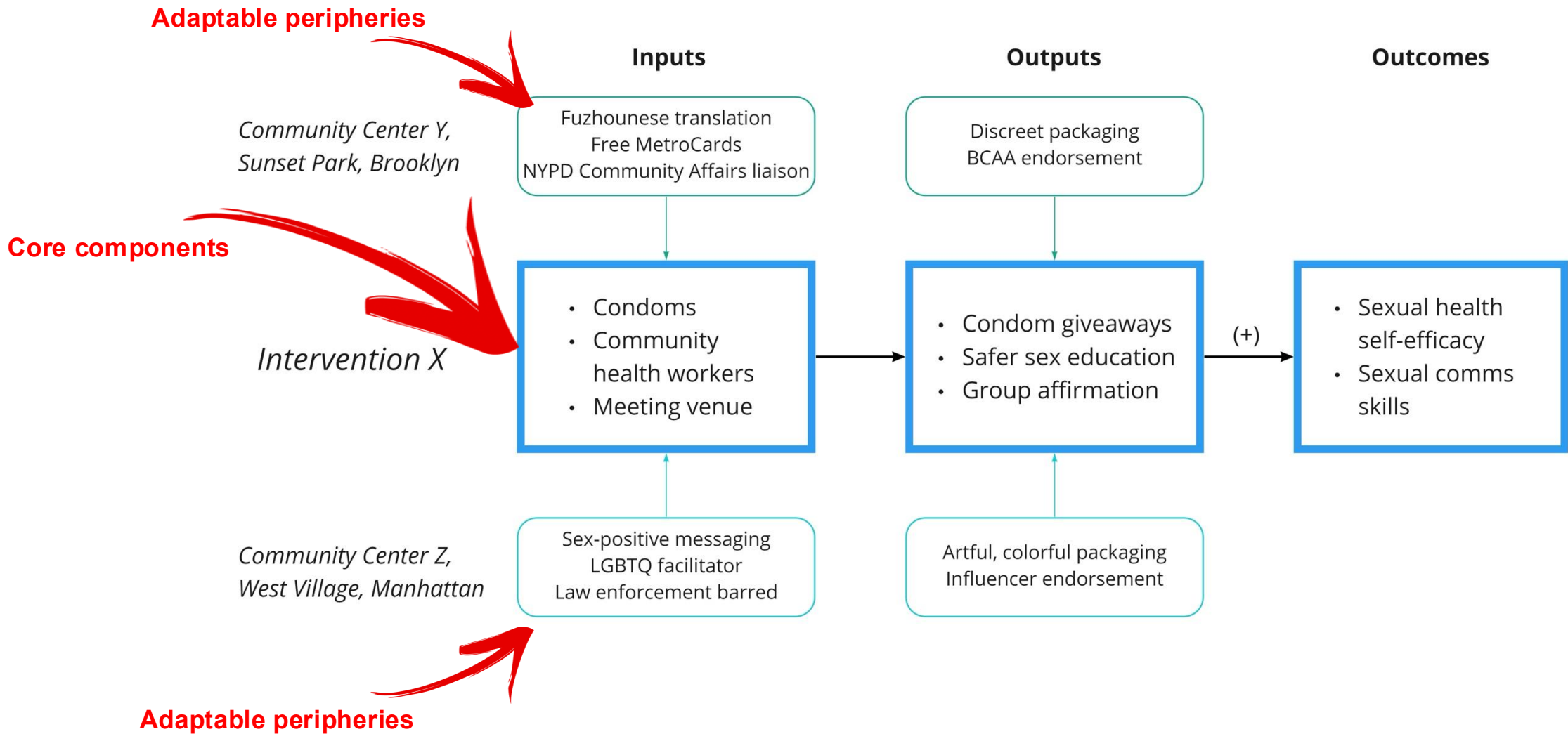
Core Components

The essential and indispensable elements of the intervention itself
(Damschroder et al., 2009).



Adaptable Peripheries

Adaptable elements, structures and systems related to the intervention and the setting into which it is being implemented.



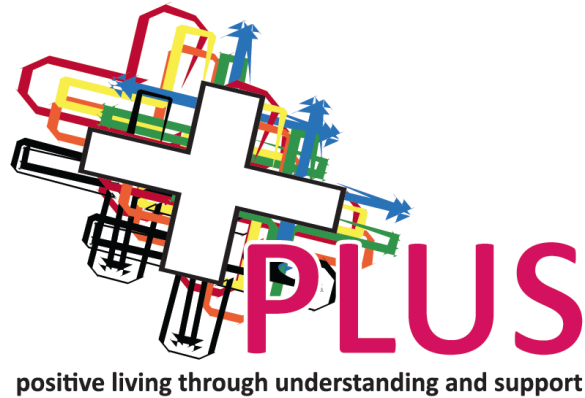
The CFIR subsumes 39 constructs across 5 domains – too many to cover here.

Let's focus on an example.

CFIR domains

- I. Intervention Characteristics
- II. Outer Setting
- III. Inner Setting
- IV. Characteristics of Individuals
- V. Process

(Damschroder et al., 2009)



Project PLUS showed promise when trialed in an academic medical center at CUNY.

Motivational Interviewing and Cognitive-Behavioral Intervention to Improve HIV Medication Adherence Among Hazardous Drinkers:

A Randomized Controlled Trial

Jeffrey T. Parsons, PhD^{*,†,‡}, Sarit A. Golub, PhD^{*,‡,§}, Elana Rosof, PhD^{*}, and Catherine Holder, BA^{*}

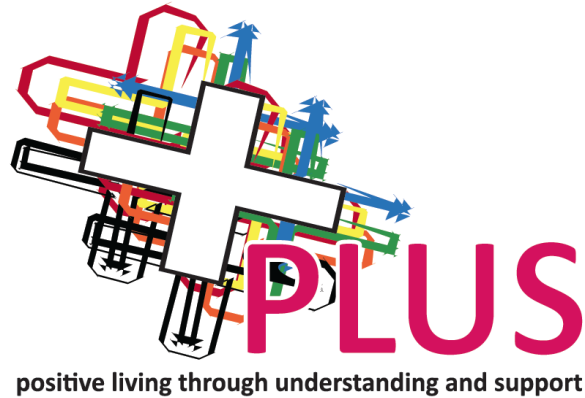
**Center for HIV/AIDS Educational Studies and Training, New York, NY*

†Department of Psychology, Hunter College, City University of New York, New York, NY

‡Social-Personality Subprogram, Graduate Center of the City University of New York, New York, NY

§Department of Psychology, Queens College of the City University of New York, Flushing, NY

condition. At the 3-month follow-up, participants in the intervention condition demonstrated significantly greater decreases in viral load and increases in CD4 cell count and self-reported adherence behavior compared with participants in the comparison condition. Project PLUS is the first behavioral adherence intervention to demonstrate such improvements in all 3 measures (viral load, CD4 cell count, and percent adherence) and is the first intervention for HIV-positive individuals with alcohol-related problems to demonstrate any significant effects.



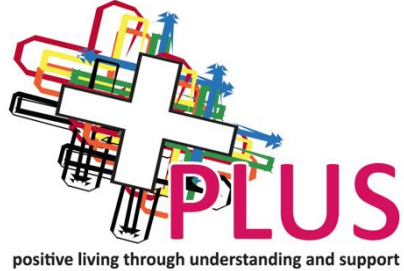
That promise was *not* realized when PLUS was implemented at three outpatient clinics in NYC.

ORIGINAL PAPER

Effectiveness of a Combined Motivational Interviewing and Cognitive Behavioral Intervention to Reduce Substance Use and Improve HIV-Related Immune Functioning

Tyrel J. Starks^{1,2}  · Simone J. Skeen³ · S. Scott Jones¹ · Sitaji Gurung¹ · Brett M. Millar¹ · Christopher Ferraris⁴ · Ana Ventuneac⁵ · Jeffrey T. Parsons⁶ · Martha A. Sparks⁷

Across outcomes, no hypothesized treatment effects were observed for the Project PLUS intervention. Results of this implementation trial therefore provide only limited evidence



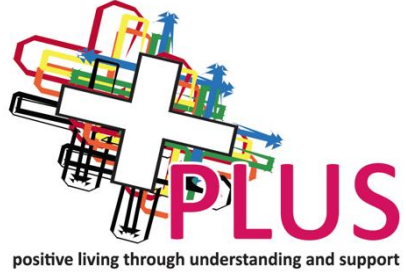
Core components

- 6 sessions MI+CBT
- reduced from 8+2 boosters
- Reduced from 1-h to 45-min

Primary outcomes

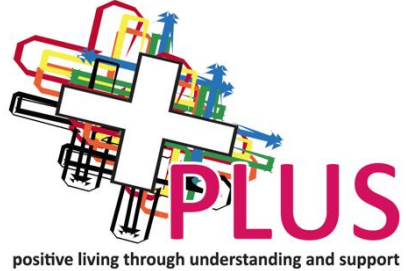
- Hazardous drug use (DAST-10)
- Hazardous alcohol use (AUDIT)
- CD4, VL (laboratory-confirmed)

Tx-outcome trajectories were uniform across conditions (intervention vs. control), across sites, across timepoints. **Why?**



PLUS was carefully adapted to suit the intervention context (Gurung et al., 2017).

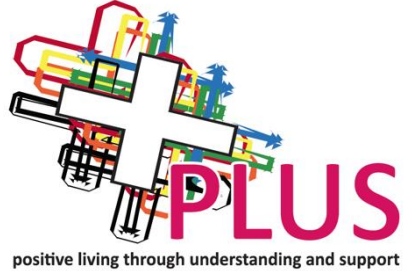
Through the lens of the CFIR, we identified possible context-specific inhibitors of Project PLUS (Starks et al., 2021).



I. Outer Setting

IA. Patient needs & resources

*“The extent to which patient needs, as well as barriers and **facilitators to meet those needs are accurately known and prioritized by the organization**” (Damschroder et al., 2009).*



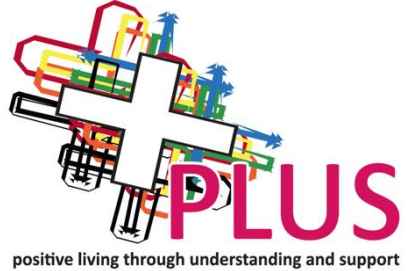
II. Outer Setting

IIA. Patient needs & resources

NYC was achieving historic successes in population HIV outcomes – meeting UNAIDS 90-90-90 Goals in 2016 (among White PLWH)^(Tian et al., 2016).

Was non-study/ “background” care under-accounted for?

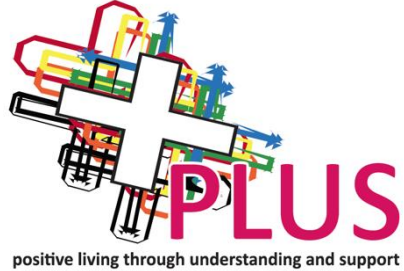
Was PLUS even necessary at that time and place?



III. Inner Setting

IIID. Implementation Climate

*“The absorptive capacity for change, **shared receptivity of involved individuals to an intervention** and the extent to which use of that intervention will be rewarded, supported, and expected within their organization.”* (Damschroder et al., 2009).

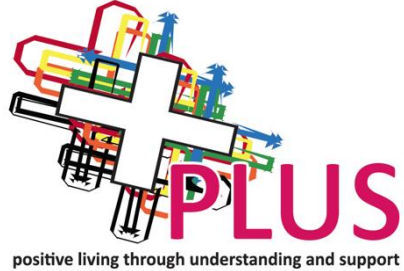


III. Inner Setting

IIID. Implementation Climate

Mt. Sinai clinicians unanimously expressed reservations regarding the time and logistical burden of implementing PLUS (Starks et al., 2021).

Did interventionists' hesitancy lead to weakened expectancy of Tx success?



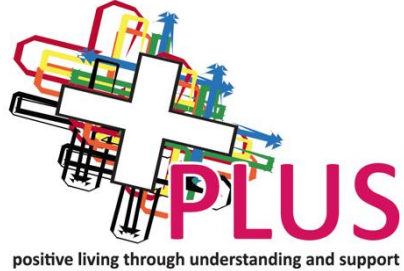
I. Intervention Characteristics

IC. Relative advantage

ID. Adaptability

“Stakeholders’ perception of the advantage of implementing the intervention versus an alternative solution.”

“The degree to which an intervention can be adapted, tailored, refined, or reinvented to meet local needs.” (Damschroder et al., 2009).



I. Intervention Characteristics

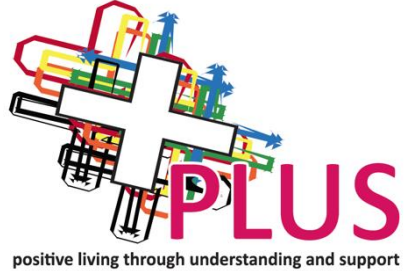
IC. Relative advantage

ID. Adaptability

Mt. Sinai clinicians were already highly skilled at delivering MI+CBT-based Tx (Starks et al., 2021).

Were interventionists unconvinced that PLUS represented an enhancement to the SoC?

Did the reduction in session number and duration undermine a *core component* of PLUS?



A final point regarding PLUS.

The original explanatory trial
launched recruitment in **July 2002**
(Parsons et al., 2007).

The outcomes of the pragmatic
(implementation) trial were first
published in **September 2021** (Starks et
al., 2021).

Is this reasonable?
Is this sustainable?

Leave the Zoom open